

University of Pretoria

SEMESTER TEST: 20 MAY 2024

COURSE: Computer Science
PAPER: COS332

TIME: 90 minutes
MARKS: 50

EXAMINER: Prof MS Olivier

THIS PAPER CONSISTS OF 11 PAGES.

ANSWER ALL QUESTIONS.

CASIO FX-82 (OR EQUIVALENT) CALCULATORS PERMITTED, AS WELL AS BASIC 5-FUNCTION CALCULATORS.

USE YOUR TEST BOOK FOR ROUGH WORK; NOTE THAT IT WILL NOT BE MARKED, BUT SHOULD BE HANDED IN.

YOU ARE ALLOWED TO WRITE ON THIS TEST PAPER.

Question 1

In each case select the alternative that fits the question best and write only the corresponding letter on your answer sheet.

- a) Which protocol will a discless computer typically use to transfer an operating system image from a server to the computer, when the computer boots?

A: DHCP
B: FTP
☒ C: TFTP
D: HTTP
E: BOOTP

- b) Which of the following are examples of network management protocols?

A: SNMP
B: CMIP
C: CMOT
D: More than one of the above
☒ E: All of the above

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- c) Which of the following statements about X.509 is/are true?

A: It is part of the X.500 series of standards introduced by CCITT.
B: It is used for public key cryptography.
C: It is used in the TCP/IP context.
D: More than one of the above
☒ E: All of the above

- d) Item which of the following protocols is an / are examples of an external gateway protocol (EGP)?

☒ A: BGP
B: OSPF
C: RIP
D: More than one of the above
E: All of the above

- e) In which of the following algorithms is *counting to infinity* a cause for concern?

A: Dijkstra
☒ B: Bellman-Ford
C: RIP
D: More than one of the above
E: All of the above

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- f) Consider a scenario in RIP where a router r sends its routing table to a router s . An entry in r 's table indicates that it can reach a network n at a cost of 5. Before receiving the message, the routing table at s indicated that it could reach n at a cost of 8. What will the cost from s to n be according to the routing table at s after s processed the message from r ?

A: 5
B: 6
C: 7
D: 8

E: More information about r is required to answer the question.

- g) Suppose computers A and B both use extended ASCII to represent characters. There is no character conversion mechanism between A and B. Assume that every byte transmitted will be received correctly; no data loss or transmission errors will occur. What challenge(s) may be experienced when these two computers communicate?

A: When A sends a digit or one of the 26 Latin characters (in either lower or uppercase) the recipient may interpret at least one of these 62 characters incorrectly.

B: A may mark up a character (for example, `ë`) and such characters may be received incorrectly at the destination.

C: It is possible that at least one of the following seven punctuation marks may not be communicated correctly: `?` `!` `,` `.` `"` `%` `:` `&`.

D: These computers will not experience any challenges since they use the same character encoding.

E: None of the problems listed above will occur (but other challenges may exist).

- h) How many Unicode characters can, in principle, be represented in UTF-8 using exactly four bytes? If the answer is expressed in the form 2^n , what is the value of n ?

A: 14
B: 20
C: 31
D: 22
E: 27

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- i) Suppose an email consists of a message encoded in HTML and a JPG image (both included in the message). These two items are the only components of the email message. Which MIME type will be used to describe the *entire* email message?

A: application/email

B: application/rfc822

C: multipart/alternative

D: multipart/mixed

E: text/html

- j) Which Content-Transfer-Encoding would be most appropriate to transfer a PNG image via SMTP?

A: 8bit

B: binary

C: quoted-printable

D: base64

E: 7bit

- k) On which ISO OSI layer should encryption be placed (if encryption is desired)?

A: 7

B: 6

C: 4

D: 3

E: It depends on factors not mentioned in the question. All of the layers mentioned above are options.

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l) Which ISO OSI layer may provide checkpoints, such that only messages sent since the last checkpoint have to be retransmitted after a connection has been lost?

- A: 2
- B: 3
- C: 4
- ☒ D: 5
- E: 6

m) How are the SYN and ACK packets sent during a TCP handshake differentiated from one another?

- ☒ A: The flags are used to determine the type of handshake packet.
- B: These packets carry a phantom byte in which the type (SYN or ACK) is encoded.
- C: The sequence and acknowledgement numbers determine the type of packet; a sequence number of 0, for example, indicates that it is a SYN packet and a sequence number of 0 and an acknowledgement number of 1 indicate that it is a SYN+ACK packet.
- D: More than one of the above
- E: All of the above

n) How does the server differentiate between the ACK packet that it receives as the last part of the handshake to establish a connection, and an ACK packet that acknowledges data that has been received?

- A: The handshake ACK packet will *always* be received before the first data acknowledgement packet.
- B: The ACK flag is only set for the handshake packet; the acknowledgement field is used to acknowledge data.
- C: For a handshake packet, the SYN or FIN flag will be set; for packets that acknowledge data neither will be set.
- D: The acknowledgement packet that acknowledges the server's SYN+ACK packet is *always* part of the handshake; packets that acknowledge other data are not.

☒ E: None of the above

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o) What is the minimum number of timer types required by any sliding window ARQ protocol?

- A: 1
- ☒ B: 2
- C: 3
- D: 4
- E: 5

p) The TCP *quiet* timer determines when a port may be used for a new TCP connection after a previous connection has been closed. Which of the following claims is/are true?

- A: The port (or socket) is in the TIME_WAIT state while the quiet timer is active.
- B: The quiet timer is activated when the FIN+ACK message is sent.
- C: The quiet timer solves the problem of a packet from a previous connection to be deemed part of a new connection.
- ☒ D: More than one of the above
- E: All of the above

q) Which command enables one to see the status of transport layer connections on a host (in most operating systems)?

- ☒ A: netstat
- B: ping
- C: tracert / traceroute
- D: tcp-show
- E: ps

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r) What type of port is port 55555?

- A: Registered
- B: Well-known
- ☒ C: Dynamic
- D: Practical
- E: Temporary

s) What does the claim that QUIC is a *quick* protocol mean?

- A: It manages the lower layers to transmit raw data at higher bit rates
- B: It reduces latency when establishing or re-establishing a connection.
- C: Where multiple parts of a message has to be transported it ensures that parts of the message are delivered quickly even if some parts are delayed.
- ☒ D: More than one of the above
- E: All of the above

t) Which of the following actions always happen(s) when a client establishes a QUIC connection with a server?

- A: The client chooses a connection ID to be used by the server.
- B: The client includes initial data encrypted with the server's public key.
- C: The client informs the server which cryptographic suite will be used.
- D: The client sends an initial packet numbered 0 on the wire.
- ☒ E: None of the above

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u) QUIC connections can migrate. Which party can initiate such a *migration* in QUICv1 or QUICv2?

- ☒ A: The client
- B: The server
- C: The network management system
- D: More than one of the above
- E: All of the above

v) Which of the following headers are always present in a long QUIC header? Include cases where the length of the field will be present and may be 0, meaning that the field itself may be omitted.

- A: Version
- B: Source connection ID
- C: Destination connection ID
- D: More than one of the above
- ☒ E: All of the above

w) Assume a client establishes a new QUIC connection at time t_0 with a server. The server receives the initial packet at time t_1 and immediately sends its response(s), which arrive(s) at time t_2 . This process continues with the client transmitting at even times ($t_0, t_2, t_4, \dots$) and the server transmitting at odd times ($t_1, t_3, t_5, \dots$). What is the soonest time at which application data may (sometimes) be sent?

- ☒ A: t_0
- B: t_1
- C: t_2
- D: t_3
- E: It is impossible to say.

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- x) Assume a client establishes a new QUIC connection at time t_0 with a server. The server receives the initial packet at time t_1 and immediately sends its response(s), which arrive(s) at time t_2 . This process continues with the client transmitting at even times ($t_0, t_2, t_4, \dots$) and the server transmitting at odd times ($t_1, t_3, t_5, \dots$). What is the soonest time at which 1-RTT application data may (sometimes) be sent?

A: t_0
☒ B: t_1
 C: t_2
 D: t_3
 E: None of the above

- y) Assume a client establishes a new QUIC connection at time t_0 with a server. The server receives the initial packet at time t_1 and immediately sends its response(s), which arrive(s) at time t_2 . This process continues with the client transmitting at even times ($t_0, t_2, t_4, \dots$) and the server transmitting at odd times ($t_1, t_3, t_5, \dots$). What is the soonest time at which the connection may be deemed to be fully established?

A: t_0
 B: t_1
 C: t_2
 D: t_3
☒ E: None of the above

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Question 2

You receive a sequence of bytes that are supposed to be UTF-8 encoded. However, some bytes have been lost during transmission. You are expected to recover all the characters that can be extracted. Where a sequence of one or more bytes does not constitute a character, you have to indicate it using an X. Use the form U+xxxx to represent the Unicode characters. Suppose you find five subsequences in the sequence where subsequence 1, 2 and 4 are valid Unicode characters, but subsequence 3 and 5 are not, then your answer may look as follows: 1) U+0001 2) U+0002 3) X 4) U+0009 5) X. Note that more spaces may have been provided on the answer sheet than actual subsequences of characters.

The byte sequence to process is the following:

```
41 D7 90 EF BA 8C
91 D0 90 D0 F0 9F
98 80 F0 90 A4 80
F0 90 8E A0 5A 87
```

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Question 3

Node A is busy communicating with node B via a TCP connection. At some time t the next byte that A will send is byte number 300. The next byte that B will send is byte number 500. The window size at A is 1000. The window size at B is 100. At time t , both A and B have successfully received all previous messages sent by its counterpart.

At time $t + 1$ A wants to send 300 bytes to node B. Call the message that A actually sends m_1 .

At time $t + 2$ B consumes 200 bytes from its buffer.

At time $t + 3$ B sends a message containing 50 bytes to A. Let us call this message m_3 .

At time $t + 4$ A sends a message to B to acknowledge receipt of m_3 . A also includes all residual data it may have to send. Let us call this message m_4 .

At time $t + 5$ B sends an empty acknowledgement message to A to acknowledge m_4 . Let us call this acknowledgement message m_5 .

- What is the sequence number included with m_1 ? (1)
- What is the acknowledgement number included with m_4 ? (1)
- What is the length of m_4 ? (1)
- What is the window advertisement included with m_4 ? (1)
- What is the window advertisement included with m_5 ? (1)

[5]

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Question 4

Briefly define or describe the following terms:

- a) TCP slow start
- b) TCP acknowledgement timer
- c) Round trip time
- d) Middlebox (especially in the context of layer 4)
- e) Protocol ossification

[5]

Question 5

Assume that the costs between nodes of a network are as follows:

A-E: 2 D-E: 5
B-E: 1 D-F: 1
B-G: 1 E-F: 10
C-F: 5

Use Dijkstra's algorithm to determine the routes from A.

A

Use the notation used in this module to show your calculations (in the table provided on the answer sheet). [10]

TOTAL

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END OF PAPER